Unsupervised Detection of Genesis Outliers in Mars HiRISE Orbital Imagery with a Two-Stage Residual Autoencoder and Latent-Space Isolation Forest

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***Abstract*—We present an end-to-end, fully unsupervised pipeline that finds tampered or foreign crops ("Genesis Outliers": spliced terrain, terrestrial content and synthetic sensor artefacts) among 10,422 HiRISE crops of 227×227 pixels, without labels and without pretrained networks. A convolutional autoencoder trained from scratch compresses every crop into a 256-d scene code; a second autoencoder compresses the first one's reconstruction residual into a 64-d code, and the concatenated 320-d latent is scored by an Isolation Forest. A robust modified z-score rule on the observed score distribution flags 48 crops (0.46 %). Two engineering steps were decisive: masking map-projection no-data wedges, which in the first two model versions had made about 96 % of all flags border crops, and the residual code, which raised the mean AUROC on a benchmark of 2,400 planted splices, glitches and man-made textures from 0.735 to 0.775 (raw-pixel Isolation Forest: 0.45). Four of the six highest-scoring crops are Earth-like aerial images; reconstruction-error heatmaps separate their diffuse, whole-frame failure from the concentrated, relief-following error of rare genuine terrain, and a metadata join attributes much of the remaining tail to two north-polar observations.**

***Index Terms*—anomaly detection, convolutional autoencoder, isolation forest, HiRISE, Mars, unsupervised learning, novelty detection, interpretability**

I. Introduction

The HiRISE camera on the Mars Reconnaissance Orbiter has produced one of the richest image archives of another planet [1]. Rare frames that do not belong must be found without anyone labelling them, which mirrors the workflow of a planetary-imaging pipeline. The competition dataset [2] contains an undisclosed number of injected outliers of three classes (*semantic splicing*, *domain-shifted terrestrial content* and *synthetic sensor artefacts*) at unknown positions.

Our contributions are: (i) a from-scratch convolutional autoencoder (CAE) with a masked structural loss and a second-stage residual code; (ii) an Isolation Forest (IF) novelty engine on the 1-D latent vectors with a statistically justified, count-free threshold; (iii) a metadata-grounded location analysis; (iv) reconstruction-error heatmaps with quantitative descriptors; (v) a documented iteration journal of six versions; and (vi) a planted-fakes benchmark that compares models without touching the hidden labels or tuning the threshold. Table I summarises the final answers.

TABLE I
Final Answers

Final model	v6: two-stage CAE trained from scratch; a fixed-length 320-dimensional latent vector (a 256-d scene code from stage 1 concatenated with a 64-d code of its reconstruction residual)
Loss	stage 1: $0.84(1-\mathrm{SSIM}) + 0.16L_1 + 0.1L_\nabla$; stage 2: MSE $+ 0.5\,L_1$; no-data masked
Novelty Score	IF $s(x, n)$ on the latent, 500 trees $\times$ 5 seeds; higher = more anomalous
Threshold	modified $z > 3.5 \Rightarrow \tau = \mathbf{0.5251}$
Flagged	**48** of 10,422 crops (0.46 %)
Top-5 (Phase 3)	06476, 06963, 07971, 03383, 07701
Location	2 enriched observations, both polar
Fusion	54 vs 48 flagged (Jaccard 0.82)

II. Data and Pre-processing

The dataset holds 10,422 single-channel 227×227 JPEG crops from 172 source observations; every crop maps to exactly one `source_image_id` with metadata (latitude, longitude, sun angle, season, resolution). Pixel values span 0–255 and are rescaled to $[0, 1]$. A sanity pass revealed that **19.7 % of crops contain a solid-black no-data wedge**, the map-projected edge of the HiRISE strip (Fig. 1): normal data, but the highest-contrast edges in the dataset.

From v3 onwards each crop receives a no-data mask: near-black (≤ 5) regions that survive a 5×5 morphological opening, touch the crop border and cover at least 0.5 % of the crop, dilated by 2 px for JPEG ringing. The opening deliberately keeps thin interior dark lines, which may themselves be injected artefacts. Masked pixels are filled with the crop's valid-pixel median and excluded from every loss term, metric and heatmap.

III. Method

A. Stage 1: Convolutional Autoencoder

The encoder applies five stride-2 3×3 convolutions (32, 64, 128, 256, 256 channels, BatchNorm, LeakyReLU), mapping $227 \to 114 \to 57 \to 29 \to 15 \to 8$; a linear layer maps the flattened $8 \times 8 \times 256$ map to the scene code $\mathbf{z}_1 \in \mathbb{R}^{256}$. The decoder mirrors it with transposed convolutions to 256×256, centre-cropped to 227×227 with a sigmoid output (11.1 M parameters). All networks are randomly initialised and trained

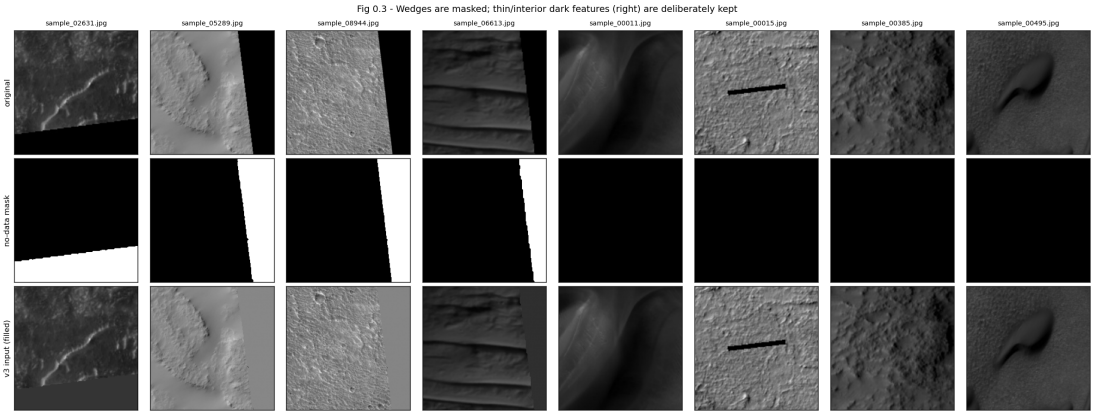


Fig. 1. No-data masking. Top: originals; middle: mask; bottom: model input. Projection wedges (left) are masked; thin interior dark features (right) are deliberately kept.

only on the competition crops (AdamW, one-cycle learning rate 10^{-3}, batch 64, 30 epochs, flip augmentation, bf16 autocast, seed 42) [3], [4].

B. Reconstruction Loss

With a validity map w (0 on no-data pixels) and $\langle\cdot\rangle_w$ the w-weighted mean,

$$\mathcal{L}_1 = 0.84\big(1 - \langle \mathrm{SSIM}(x,\hat{x})\rangle_w\big) + 0.16\,\langle |x-\hat{x}|\rangle_w + 0.1\,\mathcal{L}_\nabla, \tag{1}$$

where SSIM [5] uses an 11×11 Gaussian window ($\sigma = 1.5$), the 0.84/0.16 split follows [6], and $\mathcal{L}_\nabla$ is the L_1 distance between finite differences. SSIM penalises lost local structure that MSE averages away, L_1 anchors absolute albedo and the gradient term protects ridgelines and crater rims. No perceptual loss is used: any perceptual network would be pretrained.

C. Stage 2: Residual Code

A second autoencoder of the same design with a 64-d bottleneck is trained (MSE $+\,0.5\,L_1$ on valid pixels) on stage 1's amplified signed residual $r = 128 + 127\,\mathrm{clip}(4(x-\hat{x}),-1,1)$. The final latent is $\mathbf{z} = [\mathbf{z}_1;\mathbf{z}_2] \in \mathbb{R}^{320}$: $\mathbf{z}_1$ describes *what kind of terrain* a crop shows, $\mathbf{z}_2$ *what stage 1 could not explain.*

D. Latent Dimensionality

We trained stage 1 with $d \in \{64, 128, 256, 512\}$ for 12 epochs (Table II). Fidelity saturates (SSIM $+0.004$ from 256 to 512) and the number of principal components holding 90 % of latent variance grows sub-linearly: the extra coordinates of $d = 512$ are largely redundant and dilute the IF's uniformly random feature choice. The tail ratio $(q_{99.9} - \mathrm{median})/\mathrm{MAD}$ of the Novelty Scores is nearly flat across d, so it serves only as a sanity check; we choose $d = 256$ on fidelity and effective dimensionality.

E. Isolation Forest Novelty Score

An IF [7] with 500 trees and sub-sample size $\psi = 256$ is fitted on the latents of all crops; five forests with different seeds are averaged (rank correlation between independent forests 0.98). With $h(x)$ the path length and $c(\psi)$ its expected value,

$$s(x,\psi) = 2^{-\mathbb{E}[h(x)]/c(\psi)} \in (0,1]. \tag{2}$$

scikit-learn's `score_samples` returns $-s$; we negate it, so the **Novelty Score is s and higher means more anomalous** (correlation -0.98 with `decision_function`). `contamination` stays at `"auto"`; it only affects `predict()`, which is never called, so the anomaly count is independent of any assumed contamination fraction.

TABLE II
STAGE-1 LATENT-SIZE SWEEP (12-EPOCH SCHEDULE, SINGLE SEED)

d	SSIM	PCs (90 %)	tail ratio
64	0.682	30	10.89
128	0.678	8	9.51
256	0.692	85	11.05
512	0.696	109	10.49

F. Statistical Threshold

The final scores are not Gaussian (skewness 0.75, excess kurtosis 0.65, D'Agostino $K^2 = 885$, $p \approx 0$ [9]), and μ, σ are inflated by the very tail being sought. We use the modified z-score of Iglewicz and Hoaglin [8],

$$M_i = 0.6745\,\frac{s_i - \tilde{s}}{\mathrm{MAD}}, \qquad \text{flag } M_i > 3.5, \tag{3}$$

whose median $\tilde{s}$ and MAD have a 50 % breakdown point, so the outliers cannot move the boundary. Under a Gaussian bulk with this scale about 2.4 of 10,422 normal crops would exceed it by chance; the real bulk is right-skewed, so this is a lower bound. Table III compares the alternatives. Because the distribution is unimodal, the two-component GMM splits the skewed bulk itself and the Kneedle elbow [10] of the whole sorted curve marks the onset of the ordinary right skew; both fall inside the body of the distribution. The robust rule (48) lies between the classical $\mu + 3\sigma$ cross-check (82) and the same rule applied to log-scores (22, skew 0.54); all three sit in the

TABLE III
CANDIDATE THRESHOLDS ON THE FINAL NOVELTY SCORES

Method	Threshold	Flagged
Modified z-score	0.5251	48
Kneedle elbow of sorted scores	0.4633	987
2-component GMM posterior = 0.5	0.4291	3740
mu + 3 sigma	0.5145	82

TABLE IV
MODEL VERSIONS: PRINCIPAL COMPONENTS FOR 90 % OF LATENT VARIANCE, SPEARMAN ρ OF NOVELTY WITH NO-DATA FRACTION, CROPS FLAGGED, AND MEAN PLANTED-FAKES AUROC

Ver.	Change	PCs	ρ_{nd}	Flags	AUROC
v1	Baseline CAE, MSE	44	0.63	1596	0.497
v2	SSIM + L1 + gradient loss	36	0.66	1985	0.504
v3	No-data masking + fill	190	0.11	86	0.735
v4	Spatial $8\times8\times4$ bottleneck	139	0.07	510	0.638
v5	Photometric normalisation	187	-0.37	0	0.395
v6	Two-stage residual AE	238	0.10	48	0.775

final steep rise of the sorted scores (Fig. 2). For $k = 3, 3.5, 4, 5$ the count is 97, 48, 24, 1; k is reported, not tuned.

IV. ENGINEERING ITERATIONS

Every version uses identical data, schedule, seed and IF (Table IV; full symptom $\rightarrow$ diagnosis $\rightarrow$ fix journal in `CHANGELOG.md`). **v1 (baseline, MSE).** Reconstructions were blurry and 96 % of flags were crops with a no-data wedge (ρ(novelty, no-data fraction) $= 0.63$, 1,596 flags). **v2 (structural loss)** barely changed held-out SSIM (0.7053→0.7065): a 256-float code cannot store the phase of a ripple field, so the loss was not the binding constraint, and its edge-sensitive terms made the border problem worse. **v3 (no-data masking)** removed the nuisance mode ($\rho = 0.11$), gave a unimodal score distribution and brought Earth imagery into the flagged set. **v4 (spatial $8 \times 8 \times 4$ bottleneck)** was rejected: its coordinates encoded local brightness and it flagged 510 dark or bright scenes. **v5 (per-crop photometric normalisation)** was rejected: dividing each crop by its own contrast made planted fakes look *more* typical (benchmark AUROC 0.40). **v6 (two-stage residual code)** followed from the benchmark finding that reconstruction error alone separates fakes better than v3's latent (0.835 vs 0.735) while IF tuning did not help; it raised the benchmark AUROC to 0.775, ρ(novelty, reconstruction error) from 0.44 to 0.57, reduced the share of flags in the brightness extremes from 58 % to 44 %, and moved the Earth-like images found by eye to ranks 1, 3 and 6.

V. PLANTED-FAKES BENCHMARK

Without ground truth a model cannot be scored directly. We generate 300 instances of each of eight anomaly types from clean crops (Fig. 4: straight-seam and patch splices; bars, blocks, gain stripes and line dropouts; periodic and rectilinear man-made textures), pass them through exactly the same pre-processing as real crops, and score them with each model's IF fitted on the real data only (Table V). The threshold rule is never tuned on this benchmark; it only compares models, and its generators may differ from the organisers' injections.

TABLE V
PLANTED-FAKES BENCHMARK: AUROC OF FAKES AGAINST THE WHOLE REAL SET (LATENT + IF), RECALL AT EACH MODEL'S OWN THRESHOLD, REAL CROPS FLAGGED, AND RECONSTRUCTION-ERROR AUROC AS A REFERENCE

Model	AUROC by class: splice	glitch	domain	AUROC mean	Rec. (%)	Real flags	Recon AUROC
v1	0.58	0.51	0.39	**0.497**	4.6	1596	0.84
v2	0.51	0.56	0.39	**0.504**	3.2	1985	0.84
v3	0.61	0.72	0.88	**0.735**	9.5	86	0.84
v4	0.56	0.66	0.67	**0.638**	6.3	510	0.83
v5	0.45	0.33	0.47	**0.395**	0.0	0	0.61
v6	0.65	0.77	0.91	**0.775**	12.1	48	0.84
raw pixels	0.52	0.46	0.38	**0.453**	1.0	485	–

VI. RESULTS

A. Latent Space

PCA (50 components) followed by t-SNE [11] shows no collapse: a dense core, extended arms and small islands. Brightness is the dominant organising axis, pixel scale is partly encoded and the no-data fraction is spread uniformly, confirming the effect of masking. Flagged crops lie mainly on the periphery of the map (Fig. 5); these are visual interpretations, not labels.

B. Location Analysis

The metadata is quantised (latitude on a 5° grid plus a 90° bin; longitude only 0° or 180°E), so we analyse latitude bands (Fig. 6). Flags span 30 of 172 observations; 2 are significantly enriched (one-sided binomial test, Bonferroni $p < 0.05$): SRC_039 (9 of 16 crops, latitude bin 90°) and SRC_166 (3 of 16 crops, latitude bin 90°). Their bright, finely layered, ice-rich terrain is rare in the training distribution, so the metadata identifies these flags as a terrain class rather than tampering. The Earth-like crops, by contrast, come from four different observations at low latitudes, as expected for randomly inserted foreign content. Sun angle ($\rho = -0.03$) and |latitude| ($\rho = 0.06$) have negligible effect sizes, and flagging is independent of season (χ^2 $p = 0.53$).

C. Metadata Fusion

Concatenating the standardised latent with an encoded metadata vector (z-scored sun angle, sin/cos of latitude and longitude, $\log_2$ resolution, one-hot season) flags 54 crops against 48 (Jaccard 0.82, rank correlation 0.993). Only 10 of 330 fused dimensions are metadata, and all crops of one observation share identical metadata, so metadata can isolate an unusual observation but never a single crop: it is most useful for explaining flags after the fact.

D. Reconstruction Heatmaps and Geological Hypotheses

The five highest-scoring crops *within the thresholded set* are passed back through the stage-1 decoder (Fig. 7); error-map descriptors are reported as percentiles against 500 normal

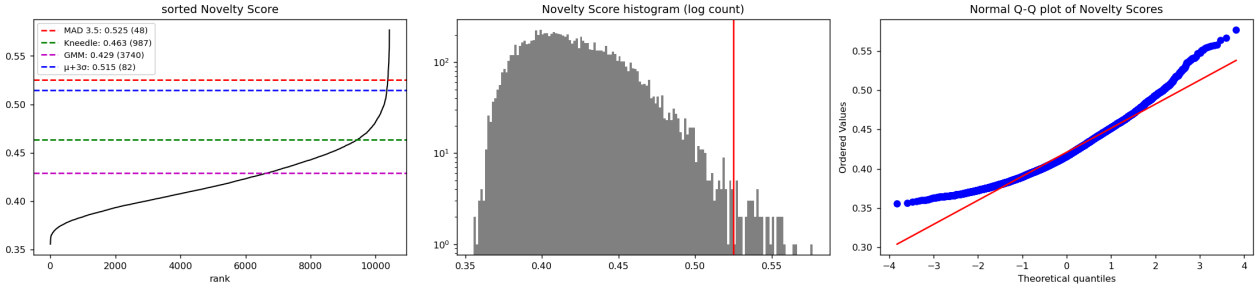


Fig. 2. Sorted Novelty Scores with candidate thresholds (left), histogram with the final threshold (centre) and normal Q–Q plot (right).

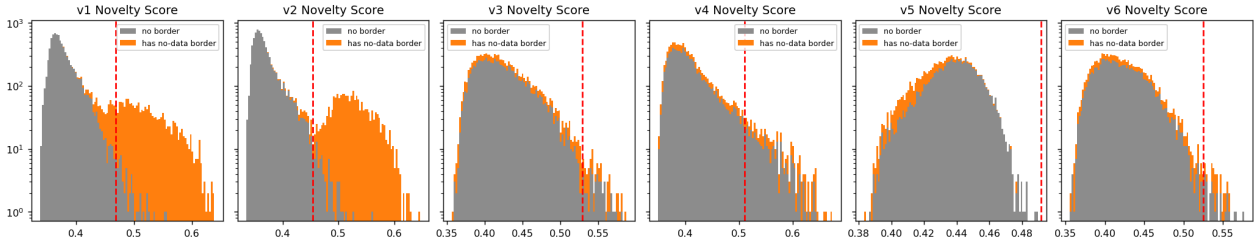


Fig. 3. Novelty Score distributions per version; orange marks crops with a no-data wedge. Dashed: modified-z threshold.

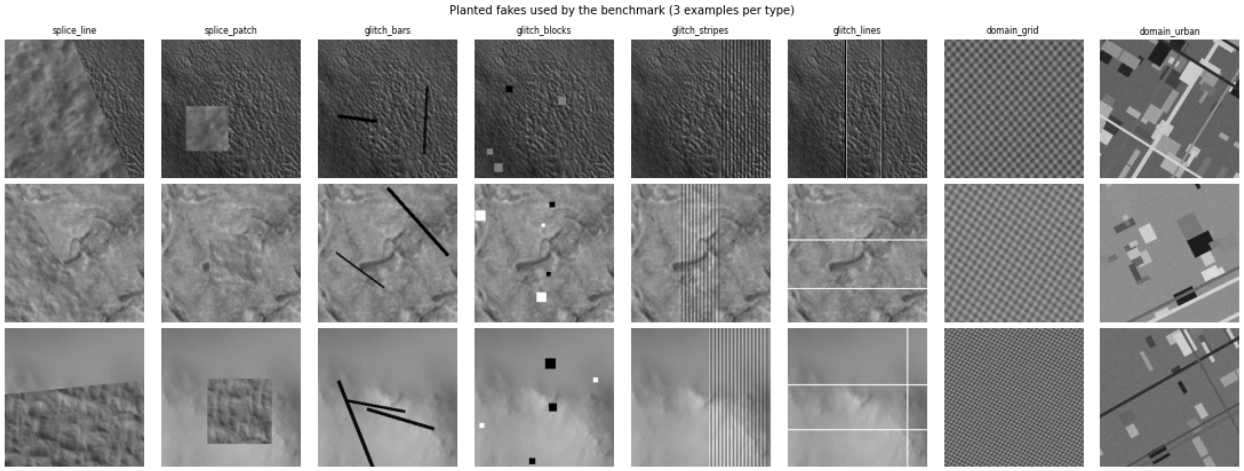


Fig. 4. Planted fakes used by the benchmark (three examples per generator).

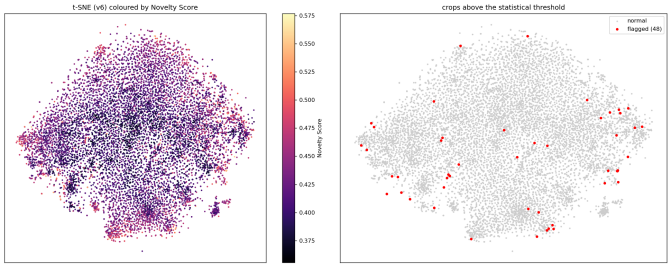


Fig. 5. t-SNE of the final latents coloured by Novelty Score (left) and crops above the statistical threshold (right).

TABLE VI
TOP-5 CROPS: NOVELTY SCORE, RECONSTRUCTION MSE AND DESCRIPTOR PERCENTILES VS. 500 NORMAL CROPS

#	Crop	Source	Nov.	MSE	Conc.	Seam
1	06476	SRC_158	0.577	0.0295	2	58
2	06963	SRC_166	0.567	0.0051	88	100
3	07971	SRC_078	0.564	0.0157	5	86
4	03383	SRC_067	0.558	0.0033	96	81
5	07701	SRC_136	0.557	0.0170	3	100

crops (Table VI). All interpretations are hypotheses. The three crops with *diffuse* error (concentration at the 2nd–5th percentile) are exactly the three with very high reconstruction error (0.016–0.030 vs. 0.003–0.005), and all three look man-made.

06476 (#1): orchard-like lattice. Straight rows of rounded,

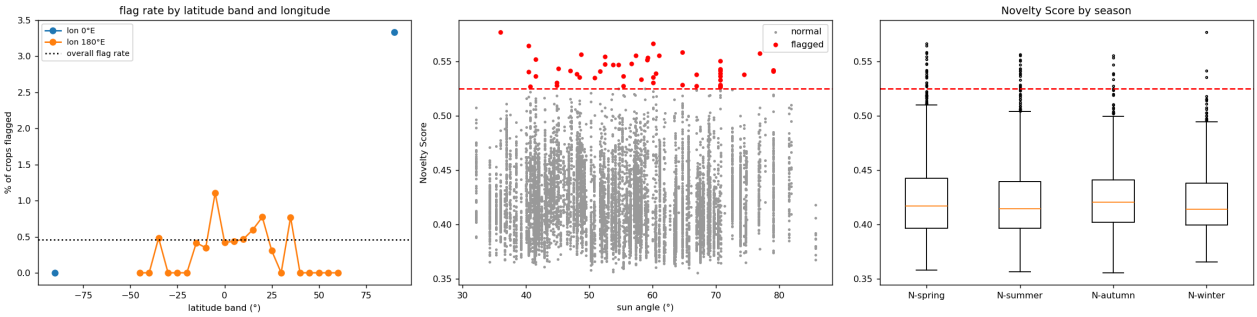


Fig. 6. Flag rate by latitude band and longitude (left), Novelty Score versus sun angle (centre) and by season (right).

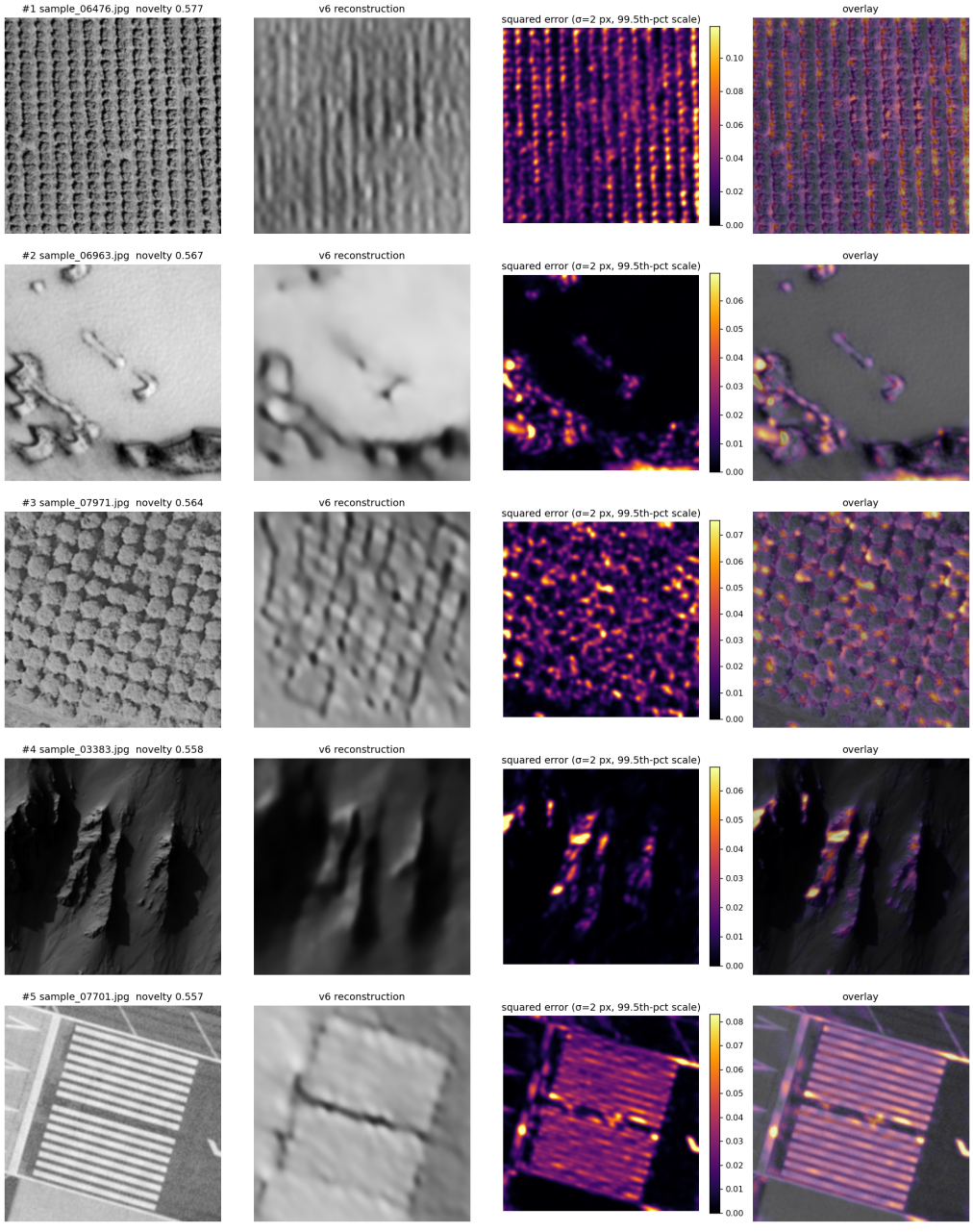


Fig. 7. Top-5 thresholded crops: original, stage-1 reconstruction, smoothed squared error (robust colour scale) and overlay.

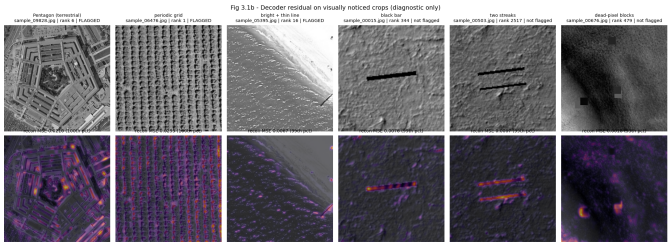


Fig. 8. Decoder residual on visually noticed crops: diffuse error on terrestrial content (left); error along the defect for synthetic glitches (right).

shadow-casting blobs, a geometry no Martian process produces at this scale; the error covers every blob (concentration at the 2nd percentile) and the reconstruction error is at the dataset's 100th percentile: a non-Martian domain shift. *06963 (#2): polar knobs.* A bright plain with dark angular knobs in the 90°N bin; error concentrated on the dark relief and low overall: rare genuine polar terrain. *07971 (#3): tree canopy.* Densely packed rounded crowns; diffuse whole-frame error and high reconstruction error, as for #1. *03383 (#4): low-sun ridges.* A very dark frame whose error sits on the thin sunlit crests: rare but Martian relief under oblique illumination. *07701 (#5): stripe markings.* Evenly spaced bright bars inside rectangles beside straight lines (runway-like painted geometry); error along every bar, high reconstruction error: foreign content, lifted into the top 5 by the residual code.

Beyond the top five (Fig. 8), the Pentagon photograph ranks 6th with diffuse error, whereas a black bar and two streaks (ranks 344 and 2,517) still lie in the top 1.5 % of reconstruction error with error along the defect: the synthetic-artefact signature, partly hidden from the latent.

VII. Limitations

Recall is incomplete: an aerial-buildings image (05508) ranks 156th, and thin streaks and dead-pixel blocks are missed by latent-space scoring. Rare genuine terrain competes with injected content in the tail. The no-data mask could hide a genuine defect that touches the crop border (e.g. a full-width dark band); v1/v2 validation metrics include no-data pixels while v3+ use valid pixels only; results are reproducible up to residual GPU non-determinism; and the planted-fakes benchmark is a proxy.

VIII. Conclusion

A diagnosed data issue (map-projection no-data) and a residual-aware latent mattered more than loss or bottleneck tweaks. The final two-stage CAE with a latent-space Isolation Forest and a robust, count-free threshold flags 48 crops; four of its top six are Earth-like images, and heatmap patterns plus metadata separate injected content from rare Martian geology. Patch-level scoring for small defects is the natural next step.

References

[1] A. S. McEwen *et al.*, "Mars Reconnaissance Orbiter's High Resolution Imaging Science Experiment (HiRISE)," *J. Geophys. Res.*, vol. 112, no. E5, 2007.

[2] G. Doran, E. Dunkel, S. Lu, and K. Wagstaff, "Mars orbital image (HiRISE) labeled data set version 3.2," Zenodo, 2020.

[3] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.

[4] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[5] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004.

[6] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, "Loss functions for image restoration with neural networks," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 47–57, 2017.

[7] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE ICDM*, 2008, pp. 413–422.

[8] B. Iglewicz and D. C. Hoaglin, *How to Detect and Handle Outliers*. Milwaukee, WI, USA: ASQC Quality Press, 1993.

[9] R. B. D'Agostino, "An omnibus test of normality for moderate and large size samples," *Biometrika*, vol. 58, no. 2, pp. 341–348, 1971.

[10] V. Satopää, J. Albrecht, D. Irwin, and B. Raghavan, "Finding a 'kneedle' in a haystack: Detecting knee points in system behavior," in *Proc. ICDCS Workshops*, 2011, pp. 166–171.

[11] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.